PA9 Surprise Drops

Fabric Mod — Software Requirements Specification

Version 1.0 · Nov 5, 2026 · Minecraft Java Edition 26.3

1. Overview

PA9 Surprise Drops is a Fabric mod for Minecraft Java Edition 26.3 that replaces every item drop in the game with a randomly assigned item in a randomly rolled quantity. When enabled, each loot table (a block, a mob, a chest, a fishing catch) is permanently assigned one replacement item, and every time that table pays out, the mod swaps the result for that item, keeping the vanilla quantity and multiplying it by a fresh random multiplier between a configurable minimum and a user-entered maximum. A multiplier of 1 keeps the normal vanilla amount.

Example: with a multiplier range of 1–1,000, breaking Stone might always drop Diamonds, but one break yields 3 Diamonds and the next yields 740. A Zombie that normally drops 2 Rotten Flesh drops 2 × the multiplier of its assigned item, so with a multiplier of 1 it drops exactly 2. A **Shuffle** button in the configuration screen re-rolls every assignment at once.

The mod is a **common** mod: all drop logic runs on the server (including the integrated server in singleplayer), and a client-side configuration screen edits the settings. This differs from PA9 TextureBuilder, which was client-only, because loot is generated server-side and cannot be changed from the client.

This specification intentionally mirrors the structure and conventions of the PA9 TextureBuilder, ShulkerPickBlock and ContainerSort specifications so that build tooling, documentation style and requirement numbering stay consistent across the author's Fabric mod projects.

Design note: assignments are derived from a single world seed through a hash of the loot table ID, rather than stored one by one. Shuffling therefore means choosing a new seed; the same seed always reproduces the same assignments. This keeps save data tiny and makes a run shareable by seed. It is a v1 design assumption and can be revisited.

2. Scope & Target Environment

Toolchain versions below are the latest stable combination found for 26.3 as of 2026-11-05, taken from published 26.3 mod release notes; confirm against fabricmc.net/develop before generating the project. "Or newer" applies to all of them.

Item	Requirement
Minecraft	Java Edition 26.3 (stable release, 2026-09-15)
Fabric Loader	0.19.5 or newer
Fabric API	0.161.0+26.3 or newer — place in .minecraft/mods/
JDK	Java 25 — required by MC 26.x. Earlier JDKs will fail.
Gradle	9.7.1+ on PATH for initial wrapper generation; gradlew is self-contained afterwards
Loom	Fabric Loom 1.18.2 — downloads the 26.3 jars on first build
Mappings	None expected — 26.x ships unobfuscated, so code uses official Mojang names (e.g. LootTable, LootContext, ItemStack). The TextureBuilder spec lists Yarn; verify what Loom 1.18.2 requires for 26.3.
IDE	IntelliJ IDEA 2025.3+ recommended
Side	Common (environment "*") — server logic on the logical server, configuration screen on the client
Optional Dep.	Mod Menu for a config shortcut. No dependency on TextureBuilder, ShulkerPickBlock or ContainerSort — see NFR-09.

Players on a dedicated server need the mod installed on the server. Clients without the mod can still join (NFR-08) but cannot open the configuration screen.

3. Project Structure & Build Setup

3.1 Source Sets

Unlike the client-only TextureBuilder, this project uses two source sets (Loom split-source layout):

- **main** — all drop logic: loot interception, seed-based assignment, multiplier rolls, world save data, commands, network receivers.
- **client** — the configuration screen, keybind, pause-menu button and the client-side network sender. Nothing in main may reference client classes.

3.2 Class Tweaker / Access Widener

Add a .accesswidener file only if a compile-time access error is actually hit (for example when reading loot-table or loot-pool internals, see 7.4). Do not pre-emptively widen access.

3.3 Build Commands (Windows / macOS / Linux)

```
# 1. Unzip and enter the project folder.
cd surprisedrops

# 2. Confirm JDK 25 is active.
java -version        # must report 25

# 3. Generate the Gradle wrapper (once only).
gradle wrapper --gradle-version 9.7.1

# 4. Build.
gradlew.bat build    # Windows
./gradlew build      # macOS / Linux

# Output:
#   build/libs/surprisedrops-1.0.0.jar          <- install this
#   build/libs/surprisedrops-1.0.0-sources.jar  <- ignore
```

On later rebuilds, step 3 is not needed — run gradlew.bat build directly.

3.4 Install

- Install Fabric Loader 0.19.5+ for Minecraft 26.3 via the Fabric installer.

- Place Fabric API 0.161.0+26.3 in .minecraft/mods/ (or the server's mods/ folder).
- Place build/libs/surprisedrops-1.0.0.jar in the same folder. On a dedicated server the mod is required on the server; it is optional on clients, which need it only to open the configuration screen.
- (Optional) Place Mod Menu in .minecraft/mods/ to open the config screen from the mod list.
- Launch the fabric-loader-26.3 profile.

4. Functional Requirements

4.1 Activation & Control

Req. ID	Requirement
FR-01	The mod SHALL have a master enabled flag, editable in the configuration screen (4.2), that turns Surprise Drops ON/OFF for the current world.
FR-02	The mod SHALL provide /surprisedrops enable and /surprisedrops disable. On a multiplayer server these require permission level 2; in singleplayer the host may always use them.
FR-03	The mod SHALL register a keybind (default: unbound) that opens the configuration screen from in-game.
FR-04	On any state change, the acting player SHALL see a brief action-bar message (e.g. "Surprise Drops: ON"). No persistent HUD overlay SHALL be rendered.
FR-05	While OFF, all drops SHALL be exactly vanilla, regardless of seed or multiplier settings.

4.2 Configuration Screen

Req. ID	Requirement
FR-06	The configuration screen SHALL be reachable from (a) a button on the vanilla pause menu, (b) the keybind in FR-03, and (c) Mod Menu if installed.
FR-07	The screen SHALL contain: the master toggle; a minimum and a maximum multiplier field; one toggle per drop category (Blocks, Mobs, Containers, Other — see FR-14); an editable world seed field; a **Shuffle** button; and Done / Cancel buttons.
FR-08	The multiplier fields SHALL accept whole numbers only, with minimum ≥ 1, maximum ≤ 10,000 and minimum ≤ maximum. An invalid entry SHALL be flagged (e.g. red text) and SHALL disable Done until fixed.
FR-09	When the maximum exceeds 1,000, the screen SHOULD show a warning that very large multipliers can create many item entities and affect performance. The player may still proceed.
FR-10	Pressing Shuffle SHALL open a confirmation prompt; only on confirmation SHALL the reassignment in 4.5 occur.
FR-11	The seed field SHALL accept a number or text (text is hashed to a number, like vanilla world seeds), so a player can reproduce a known set of assignments.
FR-12	Done SHALL send the edited settings to the server (PKT-01), where they apply immediately and are saved (FR-33). Only players with permission level 2 (or the singleplayer host) can apply changes; other players SHALL see the screen read-only.

4.3 Reassignment Mapping

Req. ID	Requirement
FR-13	The unit of reassignment SHALL be the loot table: each loot table ID (e.g. minecraft:blocks/stone, minecraft:entities/zombie, minecraft:chests/simple_dungeon) is assigned exactly one replacement item.
FR-14	Each loot table SHALL be classified into one of four categories by its loot context type: **Blocks** (block drops), **Mobs** (entity drops), **Containers** (chest, barrel, vault and other container loot), **Other** (fishing, archaeology, villager gifts, piglin bartering and all remaining types). Loot tables added by datapacks or other mods SHALL be classified the same way.
FR-15	A table's replacement item SHALL be derived as: sort the replacement pool by registry ID, hash (world seed, loot table ID) to an integer, and pick pool[hash mod pool size]. No per-table assignment is stored.
FR-16	For an unchanged seed, pool and loot table, the replacement item SHALL always be the same, across restarts and between players. Adding mods or datapacks that change the pool MAY change assignments; this is documented behaviour.
FR-17	The replacement pool SHALL be every registered item except air and a default exclusion list of operator/technical items (command blocks, structure blocks, jigsaw, barrier, light, structure void, debug stick, knowledge book). The exclusion list is editable in surprisedrops.toml (Section 6).
FR-18	Two loot tables MAY be assigned the same item, and a table MAY be assigned its own vanilla item by chance; assignment is sampling with replacement.

4.4 Multiplier & Drop Application

Req. ID	Requirement
FR-19	While ON, whenever a loot table in an enabled category generates its item list, the mod SHALL post-process that list per FR-20 to FR-23 before the items are spawned or stored.
FR-20	If the vanilla roll produced at least one item, the entire result SHALL be replaced by the table's assigned item, with a base quantity equal to the total number of items the vanilla roll produced (e.g. a Zombie dropping 2 Rotten Flesh has a base of 2). If the vanilla roll produced nothing (e.g. a leaf block that drops no sapling), nothing SHALL be added, so vanilla drop chances and Fortune/Looting effects on whether something drops are preserved.
FR-21	The multiplier M SHALL be an integer drawn uniformly from [minimum, maximum] inclusive, freshly for every drop event, using the loot context's random source. The replacement count is N = base × M, so M = 1 always gives exactly the vanilla quantity. The maximum is entered by the user (FR-07).
FR-22	N SHALL be split into stacks of the item's maximum stack size, with the remainder in a final stack (e.g. 150 Diamonds → 64 + 64 + 22).
FR-23	For items that cannot stack (maximum stack size 1), N SHALL be clamped to max_unstackable_count (default 64) to prevent mass entity spawning.
FR-24	A category whose toggle is OFF SHALL produce vanilla drops.
FR-25	Replacement stacks SHALL be created through the normal vanilla paths (item entities for drops, container slots for container loot), so pickup, hoppers, despawn timers and merging behave as normal.

Req. ID	Requirement
FR-26	For container loot, if N exceeds the container's free capacity, the excess SHALL be discarded with a debug-level log entry. The mod SHALL NOT overwrite or remove any item already in the container.
FR-27	Items that do not come from a loot table roll SHALL NEVER be altered: player death drops, items a player drops or throws, contents of a broken container, and experience orbs. Loot rolls triggered by commands (e.g. /loot) are ordinary loot rolls and ARE affected.

4.5 Shuffle

Req. ID	Requirement
FR-28	Shuffle SHALL generate a new random 64-bit world seed and apply it immediately, so every loot table is re-derived under FR-15.
FR-29	Shuffle SHALL affect future drops only. Items already dropped and containers already generated SHALL NOT change; containers whose loot has not yet generated SHALL use the new seed.
FR-30	Shuffle SHALL NOT change the multiplier range, category toggles or the master toggle.
FR-31	After a shuffle, the acting player SHALL see a message showing the new seed, so it can be copied and reused (FR-11).

4.6 Persistence & Commands

Req. ID	Requirement
FR-32	Global defaults and non-GUI options SHALL live in /config/surprisedrops.toml (Section 6). It supplies the initial values for any new world.
FR-33	Per-world state — enabled flag, seed, multiplier range and category toggles — SHALL be saved in the world's save data so each world keeps its own assignments. The first time the mod loads in a world, it SHALL generate a random seed.
FR-34	The mod SHALL provide: /surprisedrops shuffle [seed], /surprisedrops multiplier <min> <max>, /surprisedrops reload (re-reads the TOML without a restart) and /surprisedrops lookup <loot table ID> (prints the assigned item and current range).
FR-35	The same configuration screen SHALL be exposed through Mod Menu if installed (optional dependency).

5. Non-Functional Requirements

5.1 Performance

Req. ID	Requirement
NFR-01	Resolving a loot table's replacement item (FR-15) SHALL take under 0.1 ms after the replacement pool is built. The pool SHALL be built once at server start and rebuilt only on datapack reload, never per drop.
NFR-02	The mod SHALL add zero measurable overhead to loot generation while OFF or while a loot table's category is disabled (an early return before any hashing).

Req. ID	Requirement
NFR-03	A single drop event at the maximum multiplier (10,000 × the vanilla quantity) SHALL complete within one server tick without a visible lag spike on typical hardware. FR-22 and FR-23 bound the number of entities created.

5.2 Reliability & Safety

Req. ID	Requirement
NFR-04	If an exception occurs while replacing a drop, the mod SHALL catch it, log a warning, return the vanilla drops for that roll, and notify operators via chat. If three consecutive rolls fail, the mod SHALL disable itself for the rest of the session.
NFR-05	The mod SHALL NOT duplicate or delete any item a player owns. Only the output of a loot table roll is ever replaced (FR-27).
NFR-06	Assignments SHALL be fully reproducible: the same seed, pool and loot table SHALL give the same item after any restart (FR-16).

5.3 Compatibility

Req. ID	Requirement
NFR-07	The mod SHALL function in survival, creative and adventure modes and in all difficulties.
NFR-08	Clients without the mod SHALL be able to join a server that has it. Mod-specific packets SHALL be sent only to clients that have the mod, and gameplay does not depend on them.

Req. ID	Requirement
NFR-09	The mod SHALL coexist with PA9 TextureBuilder, ShulkerPickBlock and ContainerSort without conflict. None of the four mods mixin into the same target methods, and Surprise Drops depends on none of them.
NFR-10	Loot tables supplied by datapacks and other mods SHALL be handled through the same classification as vanilla ones (FR-14), and a /reload SHALL rebuild the replacement pool without a restart.

5.4 Maintainability

Req. ID	Requirement
NFR-11	Every mixin SHALL carry Javadoc explaining what is intercepted and why, following the ShulkerPickBlock/ContainerSort documentation standard.
NFR-12	Mixin targets SHALL be limited to methods declared on the target class itself, never inherited methods. Where Fabric API offers an event that does the same job, the event SHALL be used instead of a mixin.
NFR-13	Assignment, multiplier and stack-splitting logic SHALL live in a dedicated utility class (SurpriseDropsHelper) with no Minecraft event or render dependencies, so it can be unit-tested without a running game.

6. Configuration Options (surprisedrops.toml)

These are defaults for new worlds plus options that have no GUI control. Values changed in the configuration screen are stored per world (FR-33) and override the matching defaults for that world.

Option Key	Type	Default	Description
enabled_default	Boolean	true	Whether Surprise Drops starts ON in a newly created world (FR-01).
multiplier_min	Integer	1	Default minimum multiplier per drop event (FR-21); 1 means the normal vanilla amount. Range 1–10000, must be ⩽ multiplier_max.
multiplier_max	Integer	1	Maximum multiplier per drop event, entered by the user in the config file or screen. Range 1–10000. The default of 1 keeps every vanilla quantity until the user raises it; e.g. 1000 allows any multiplier from 1 to 1,000.
affect_blocks	Boolean	true	Randomize block drops (FR-14, FR-24).
affect_mobs	Boolean	true	Randomize mob/entity drops.
affect_containers	Boolean	true	Randomize chest, barrel, vault and other container loot.
affect_other	Boolean	true	Randomize fishing, archaeology, gifts, bartering and other loot.
max_unstackable_count	Integer	64	Cap on N for items with maximum stack size 1 (FR-23). Range 1–1000.
excluded_items	Array of String	technical items listed in FR-17	Item IDs that can never be chosen as a replacement (FR-17).

Option Key	Type	Default	Description
open_config_keybind	String	unbound	Keycode for opening the configuration screen (FR-03). Also rebindable in the vanilla Controls screen.
debug_logging	Boolean	false	Write assignment, multiplier and discard diagnostics to the Fabric log. Development use only.

7. Mixin & Loot Architecture (Implementation Guidance)

The strategy mirrors the other PA9 mods: inject into specific target-class methods, never target inherited methods, and keep business logic out of mixin classes. Names below use official Mojang naming and **must be verified against the 26.3 jars** before relying on them.

7.1 Drop Pipeline

Every category in FR-14 flows through the same loot-table code, so one interception point covers them all:

1. Block break, mob death, container open, fishing and similar events call the loot table to generate a list of ItemStacks.
2. The interception receives that list, the loot table ID and its loot context type.
3. SurpriseDropsHelper checks the master toggle and category toggle (FR-05, FR-24); if either blocks, the list is returned untouched.
4. If the list is empty, it is returned untouched (FR-20).
5. Otherwise the list is replaced by the assigned item (FR-15), with count = the vanilla total × a freshly rolled multiplier (FR-20, FR-21), split into stacks (FR-22, FR-23).
6. Vanilla code then spawns the item entities or fills the container as usual (FR-25).

7.2 Interception Point

- **Preferred:** Fabric API's loot events (LootTableEvents, specifically a drops-modification event if available in Fabric API 0.161 — verify). It receives the generated drops and lets the mod rewrite them without a mixin (NFR-12).

- **Fallback:** a mixin on the LootTable class (net.minecraft.world.level.storage.loot.LootTable, verify), at the return of the method that produces the item list (getRandomItems, verify). Do not target methods inherited from a superclass.
- Container loot passes through the same list-producing method before the vanilla fill code places the stacks, so overflow is handled by vanilla's own over-fill protection (FR-26). Verify this in the 26.3 source.
- Category (FR-14) is read from the loot table's context type (LootContextParamSet): block, entity, chest, and everything else → Other.

7.3 Pause-Menu Button

- Target: the vanilla pause screen class (PauseScreen, verify), injecting at the tail of its init method so the button can reference final widget positions.
- Logic: add a button that opens SurpriseDropsConfigScreen with the pause screen as parent (FR-06).

7.4 SurpriseDropsConfigScreen (Non-Mixin Custom Screen)

A standard Screen subclass in the client source set rendering the controls in FR-07. It edits a local copy of the settings, validates per FR-08, and sends them to the server on Done (FR-12). Read-only for non-operators.

Add an access widener only if reading loot-table internals is needed and produces a compile error; do not widen final fields (avoids IllegalAccessError under Loader 0.19.x).

7.5 SurpriseDropsHelper (Non-Mixin Utility)

Pure Java, no Minecraft event or render dependencies:

- buildPool(List<String> allItemIds, Set<String> excluded) → List<String> — sorted pool per FR-17.
- assignedItem(long seed, String lootTableId, List<String> pool) → String — hash-based pick per FR-15.
- rollCount(int min, int max, RandomSource random) → int — uniform draw per FR-21.
- splitStacks(int count, int maxStackSize, int unstackableCap) → int[] — stack sizes per FR-22 and FR-23.

7.6 World State

A SavedData subclass (attached to the overworld's data storage) holds enabled flag, seed, multiplier range and category toggles (FR-33). It is the single source of truth read by the interception on every roll.

8. Client–Server Synchronisation (Packet Handling)

Drops are generated and spawned entirely on the server, so the item entities reach every client through ordinary vanilla packets. Custom packets exist only to carry the configuration screen's settings and are optional for clients (NFR-08).

Req. ID	Requirement
PKT-01	Done in the configuration screen SHALL send one custom C2S payload (Fabric networking) containing the enabled flag, multiplier range, category toggles and seed.
PKT-02	The server SHALL re-check the sender's permission (FR-12) and the value ranges (FR-08) on receipt, and SHALL ignore the payload if either check fails. The server never trusts the client's validation.
PKT-03	A Shuffle request SHALL be a separate C2S payload with no seed; the server generates the new seed itself (FR-28). An explicit seed (FR-11) travels in the settings payload.
PKT-04	The server SHALL send the current settings to a player with the mod when they join and after every change, so the screen always opens with live values.
PKT-05	Payloads SHALL be registered as optional. The server SHALL send them only to clients that report they can receive them, so vanilla clients are never disconnected.

Req. ID	Requirement
PKT-06	Loot replacement SHALL use no custom packets: replaced drops spawn through the vanilla item-entity and container-update packets.

9. Suggested Mod Metadata (fabric.mod.json)

Modelled on the TextureBuilder fabric.mod.json, with versions updated for 26.3 and the environment changed to common.

```
{
  "schemaVersion": 1,
  "id": "surprisedrops",
  "version": "1.0.0",
  "name": "PA9 Surprise Drops",
  "description": "Every drop becomes a random item in a random quantity.",
  "authors": ["<Your Name>"],
  "license": "MIT",
  "environment": "*",
  "entrypoints": {
    "main": ["com.yourname.surprisedrops.SurpriseDrops"],
    "client": ["com.yourname.surprisedrops.SurpriseDropsClient"]
  },
  "mixins": ["surprisedrops.mixins.json"],
  "accessWidener": "surprisedrops.accesswidener",
  "depends": {
    "fabricloader": ">=0.19.5",
    "fabric-api": ">=0.161.0",
    "minecraft": "~26.3",
    "java": ">=25"
  },
  "suggests": {
    "modmenu": "*"
  }
}
```

Omit the accessWidener line and file if 3.2 never requires one.

10. Out of Scope

- Assignment by original item rather than by loot table (e.g. every Cobblestone drop anywhere becoming the same item) — reserved for a future version.
- A fresh random item on every drop with no stable assignment, and a fixed multiplier per assignment — v1.0 is fixed item per loot table with a multiplier rolled fresh each drop.
- Editing, locking or pinning individual assignments in the GUI, and a GUI editor for the exclusion list (TOML only in v1.0).
- Weighting the replacement pool by item rarity or progression stage.
- Altering experience orbs, player death drops, player-thrown items, broken-container contents, or any item not produced by a loot table (FR-27).
- Mob equipment and other drops that bypass loot tables.
- A persistent HUD overlay showing the seed or assignments — the screen, commands and action-bar messages are the only UI for v1.0.
- Forge, NeoForge, Quilt, or other mod loaders.
- Bedrock Edition.

11. Acceptance Criteria

All scenarios below must pass in a MC 26.3 client and server with Fabric Loader 0.19.5+ and Fabric API 0.161.0+26.3 installed before the mod is considered complete.

TC ID	Name	Setup	Expected Result
TC-01	Master toggle	Run /surprisedrops disable, break Stone, then /surprisedrops enable and break Stone again.	OFF drops Cobblestone as vanilla; ON drops the assigned item; action-bar message shown each time (FR-01, FR-02, FR-04, FR-05).
TC-02	Config screen access	Open the pause menu and click the Surprise Drops button; repeat via keybind.	Screen opens showing toggle, min/max fields, four category toggles, seed field and Shuffle (FR-06, FR-07).

TC ID	Name	Setup	Expected Result
TC-03	Stable assignment	Break 20 Stone blocks with one seed.	Every break drops the same item type; only the count varies (FR-13, FR-15, FR-16).
TC-04	Multiplier range	Set range 1–1000 and break 500 Stone blocks (vanilla drop of 1), recording each count.	All counts lie in [1, 1000], values vary, and the average is near 500 (FR-21).
TC-05	Stack splitting	Force a count of 150 of a stackable item (e.g. via a test seed and fixed range 150–150).	Three stacks of 64, 64 and 22 spawn (FR-22).
TC-06	Shuffle	Note the assigned item for Stone and Zombie, press Shuffle and confirm.	New seed is displayed; assignments change; items already dropped are unchanged; range and toggles are unchanged (FR-10, FR-28–FR-31).
TC-07	Seed reproducibility	Record assignments, enter a different seed, then re-enter the original seed.	The original assignments return exactly, including after a restart (FR-11, FR-16, NFR-06).
TC-08	Category toggle	Turn Mobs OFF, kill a Zombie, then break Stone.	Zombie drops vanilla loot; Stone is still replaced (FR-24).
TC-09	Container loot	Open an ungenerated chest in a structure.	Contents are the assigned item × N, split into stacks; nothing pre-existing is overwritten (FR-26).
TC-10	Other loot	Fish until a catch occurs; trade-gift or barter if available.	Catches and gifts are replaced by the assigned item × N (FR-14).
TC-11	Empty roll preserved	Break Oak Leaves repeatedly.	Leaves that drop nothing in vanilla still drop nothing; roughly the vanilla fraction drops the assigned item (FR-20).

TC ID	Name	Setup	Expected Result
TC-12	Unstackable clamp	Find or set a seed where a table maps to a sword or tool; break that block with range 1–1000.	Never more than max_unstackable_count items drop (FR-23).
TC-13	Player items untouched	Die with a full inventory; break a chest containing items.	Death drops and chest contents are exactly the original items (FR-27, NFR-05).
TC-14	Persistence	Set a custom range and seed, quit, relaunch; then create a second world.	First world keeps its settings; the second world has its own random seed and the TOML defaults (FR-32, FR-33).
TC-15	Permissions	On a dedicated server, join as a non-operator and open the screen; try /surprisedrops shuffle.	Screen is read-only; the command is denied; a spoofed payload is ignored (FR-12, PKT-02).
TC-16	Vanilla client	Join the modded server with an unmodded client and break blocks.	No kick or error; replaced drops appear normally (NFR-08, PKT-05).
TC-17	Invalid input	Enter min 500 and max 100.	Fields flagged and Done disabled (FR-08).
TC-18	Commands	Run /surprisedrops lookup minecraft:blocks/stone, then edit the TOML and run /surprisedrops reload.	Lookup prints the assigned item and range; reloaded options take effect without a restart (FR-34).
TC-19	Vanilla quantity kept	Set the multiplier range to 1–1 and kill Zombies until one drops 2 Rotten Flesh in vanilla terms (check with the feature off, or by count).	The Zombie drops exactly 2 of its assigned item; raising the range to 1–5 gives 2, 4, 6, 8 or 10 (FR-20, FR-21).

12. Risks & Mitigations

Risk	Level	Description	Mitigation
Mass item-entity spawning	HIGH	A single drop at multiplier 1,000+ (or an unstackable item) can create hundreds of entities and lag the server.	Stack splitting (FR-22), unstackable clamp (FR-23), hard maximum of 10,000 (FR-08), warning above 1,000 (FR-09).
Interception point changes in 26.3	MEDIUM	The loot-table method names or the Fabric API drops event may differ from this spec.	Section 7 marks every name as verify-first; prefer the Fabric event over a mixin (NFR-12); keep logic in SurpriseDropsHelper.
Progression-breaking or unobtainable replacements	MEDIUM	Random picks may give technical or creative-only items, or hand out endgame items from the first block.	Default exclusion list (FR-17), editable in TOML. Players who want a fairer run can lower the maximum multiplier.
Assignments shift when mods or datapacks change	MEDIUM	The pool is sorted by registry ID, so adding or removing items changes which item a hash selects.	Documented in FR-16; players can keep the pool stable or reshuffle with a known seed after changing mods.
Settings spoofed by non-operators	MEDIUM	A modified client could send a settings payload.	Server-side permission and range checks on every payload (PKT-02).
Container overflow	LOW	A large N can exceed a chest's capacity, discarding some items.	Never overwrite existing contents (FR-26); document behaviour; debug-log discards.
Non-loot-table drops stay vanilla	LOW	Mob equipment and similar drops bypass loot tables and are not randomized.	Listed in Out of Scope; revisit in a later version if wanted.

Risk	Level	Description	Mitigation
Naming differences in the 26.x toolchain	LOW	This spec lists no mappings layer because 26.x is unobfuscated, whereas TextureBuilder used Yarn.	Confirm with the Loom 1.18.2 template at project creation; update Section 2 if a mappings entry is needed.